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Course: Software Engineering Lab 1

Problem Statement #18: Clinical Trial Subject Recruitment Engine

Exception Flow Document

Use Case: UC-03 Check Eligibility

Primary Actor: Clinical Researcher

Secondary Actor: Trial Candidate

Note: The main success scenario for this use case is documented in the Use-Case Flow document. This document lists the exception flows only, that is the error conditions that can stop the normal flow.

Preconditions:

1. The Trial Candidate has registered an account (UC-01) and submitted the pre-screening survey (UC-02).

2. The Clinical Researcher is logged in and has access to the eligibility rule engine.

3. At least one active trial protocol with inclusion and exclusion criteria is expected to exist.

Postconditions (on exception):

1. No partial or wrong eligibility result is saved and the candidate record stays in a consistent state.

2. The system writes the exception and the reason to the audit trail.

Reference Steps of the Main Success Scenario:

1. Clinical Researcher opens the candidate list and selects a candidate.

2. System displays the submitted pre-screening survey for that candidate.

3. Researcher selects the trial protocol to check the candidate against.

4. Researcher clicks the Check Eligibility action.

5. System runs the eligibility rule engine on the survey responses.

6. System stores and displays the eligibility result.

Exception Flow E1: Incomplete Survey Data (at step 5)

E1.1. The rule engine detects that one or more mandatory survey fields are empty.

E1.2. System stops the eligibility check and does not store any result.

E1.3. System shows the message that the survey is incomplete and lists the missing fields.

E1.4. System sends the candidate a request to complete the survey (UC-02).

E1.5. Use case ends without an eligibility decision.

Exception Flow E2: Eligibility Rule Engine Unavailable (at step 5)

E2.1. System calls the rule engine but does not get a response within the timeout.

E2.2. System retries the call up to two times.

E2.3. If all retries fail, System keeps the candidate status as Pending and does not store a result.

E2.4. System shows the message that the service is not available and asks the researcher to try again later.

E2.5. System logs the failure in the audit trail. Use case ends.

Exception Flow E3: No Active Trial Protocol Configured (at step 3)

E3.1. The researcher opens the trial list but no active protocol with criteria is found.

E3.2. System disables the Check Eligibility action.

E3.3. System shows the message that no active trial is configured and asks an admin to add one.

E3.4. Use case ends without an eligibility decision.

Exception Flow E4: Duplicate Candidate Record (at step 1)

E4.1. System finds more than one record with the same candidate identity while opening the profile.

E4.2. System blocks the eligibility check to avoid a wrong result.

E4.3. System shows the message that a duplicate record exists and asks the researcher to merge or pick the correct record.

E4.4. After the researcher resolves the duplicate, the main success scenario can start again from step 1.

Exception Flow E5: Researcher Session Timeout (any step)

E5.1. The researcher session expires while the eligibility check is in progress.

E5.2. System cancels the current operation and does not save any partial result.

E5.3. System redirects the researcher to the login page.

E5.4. After a fresh login the researcher restarts the use case from step 1.